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EX NO: 15	AI-based Traffic Prediction for 5G Networks

AIM: To develop and simulate an AI-based traffic prediction model for 5G networks using

MATLAB and evaluate its performance in terms of Prediction Accuracy (%), Mean Squared

Error (MSE), and Network Throughput (Mbps).

Procedure:

1. Generate or import a dataset representing actual 5G network traffic over different time

intervals.

2. Preprocess the data by normalizing the traffic values and separating them into training

and testing datasets.

3. Develop a simple AI-based prediction model (e.g., Linear Regression or Neural

Network) to estimate future network traffic.

4. Predict the traffic values using the trained AI model and compare them with the actual

traffic values.

5. Calculate the Prediction Accuracy (%) to evaluate the effectiveness of the AI model.

6. Compute the Mean Squared Error (MSE) between the predicted and actual traffic

values to measure prediction error.

7. Estimate the Network Throughput (Mbps) based on the predicted traffic load and

available network resources.

8. Plot MATLAB graphs for Prediction Accuracy, MSE, and Network Throughput, and

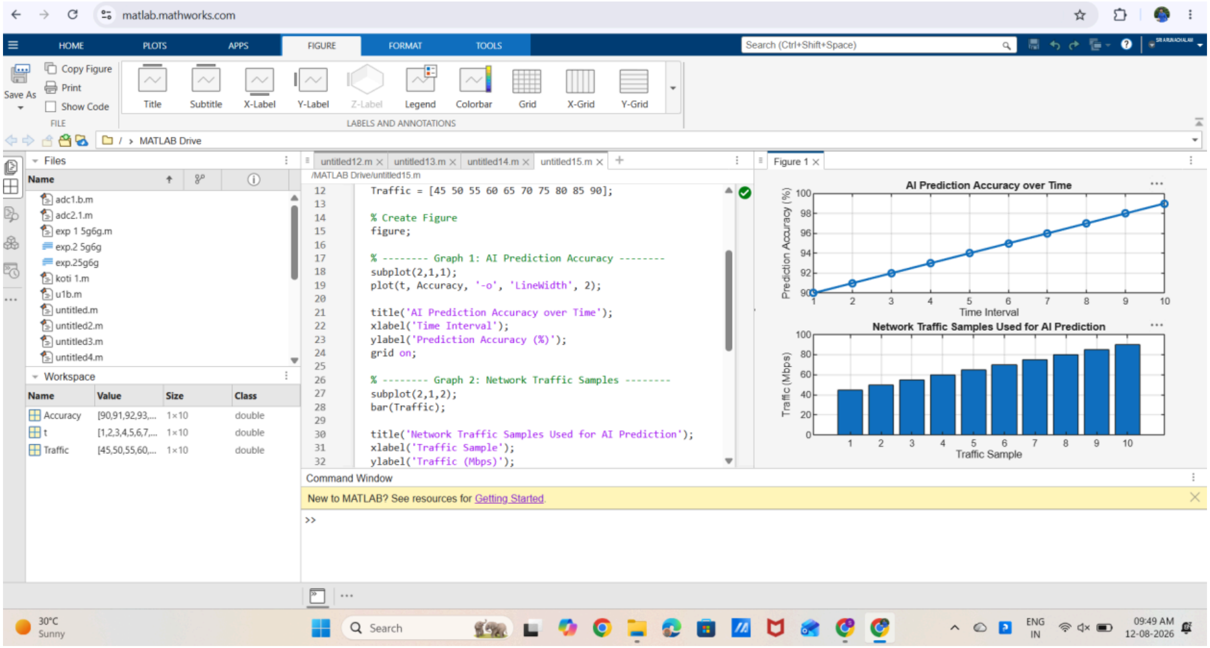
analyze the AI model's performance for efficient 5G network traffic management.

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% Time samples
t = 1:10;
% Prediction Accuracy (%)
Accuracy = [90 91 92 93 94 95 96 97 98 99];

% Actual Network Traffic (Mbps)
Traffic = [45 50 55 60 65 70 75 80 85 90];
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(t,Accuracy,'-o','LineWidth',2)
title('AI Prediction Accuracy over Time')
xlabel('Time Interval')
ylabel('Prediction Accuracy (%)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
bar(Traffic)
title('Network Traffic Samples Used for AI Prediction')
xlabel('Traffic Sample')
ylabel('Traffic (Mbps)')
grid on
```

OUTPUT:



Result:

 The first graph shows that the AI model achieves progressively higher prediction accuracy over successive time intervals.

 The second graph represents the network traffic samples used by the AI model, illustrating varying traffic loads in the 5G network.